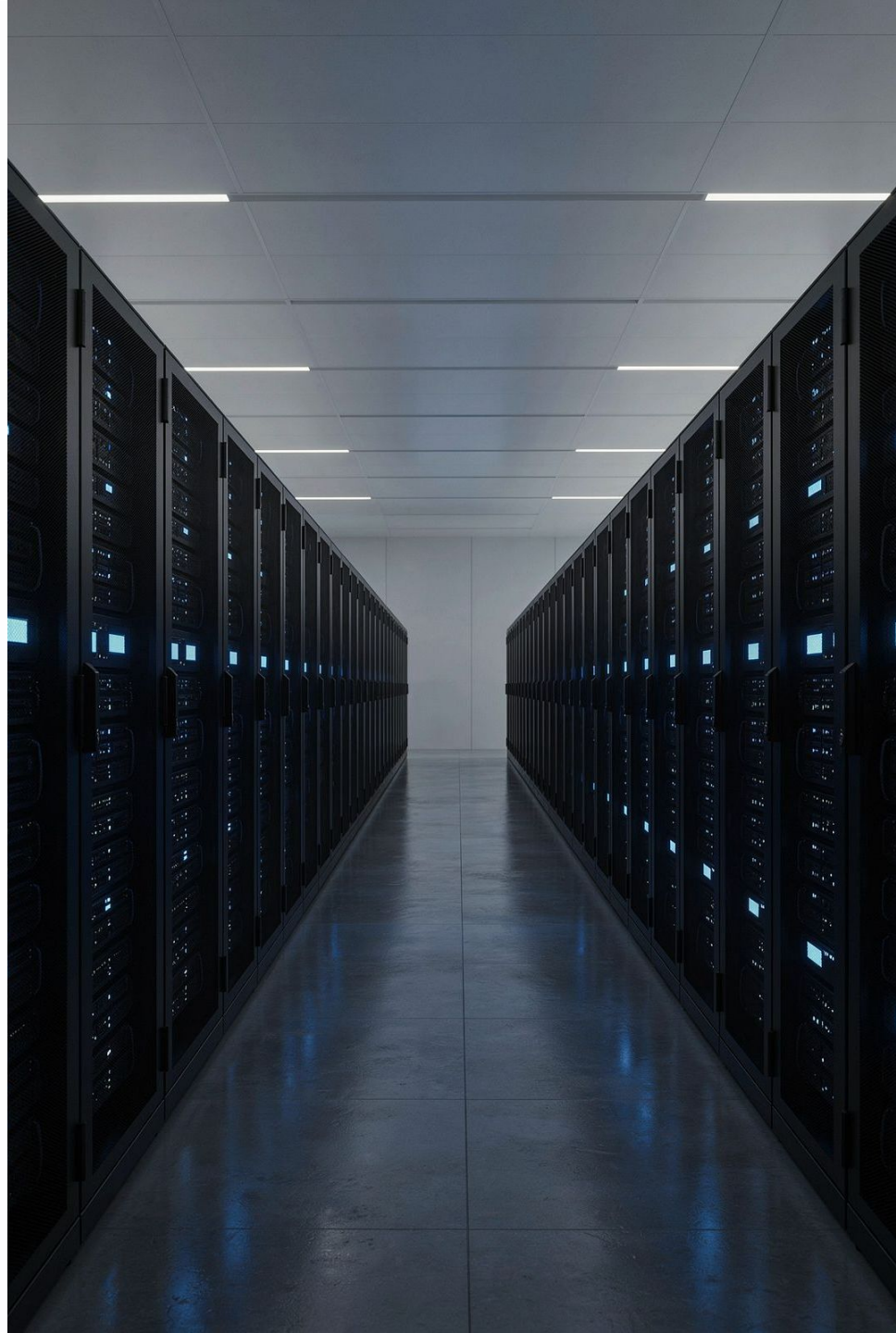
AWS AI-Powered Developer Tools

From writing infrastructure code to exploring resources and debugging logs — AWS embeds intelligent assistance directly into your everyday workflow. This presentation walks through five powerful tools that help you build faster, smarter, and with greater confidence.



OVERVIEW

Five Tools, One Goal: Move Faster on AWS

Think of these tools as having a senior AWS engineer sitting next to you at all times — one who knows every service, every CLI command, and every log pattern. Here's what we'll cover:

01

Amazon Q Developer for IaC

Generate and explain Infrastructure as Code in seconds

02

AWS Console Q Assistant

Ask questions and get guidance right inside the AWS Console

03

CLI with Q

Auto-complete and explain shell commands intelligently

04

CloudWatch Logs Insights

Query and analyze logs without writing complex syntax

05

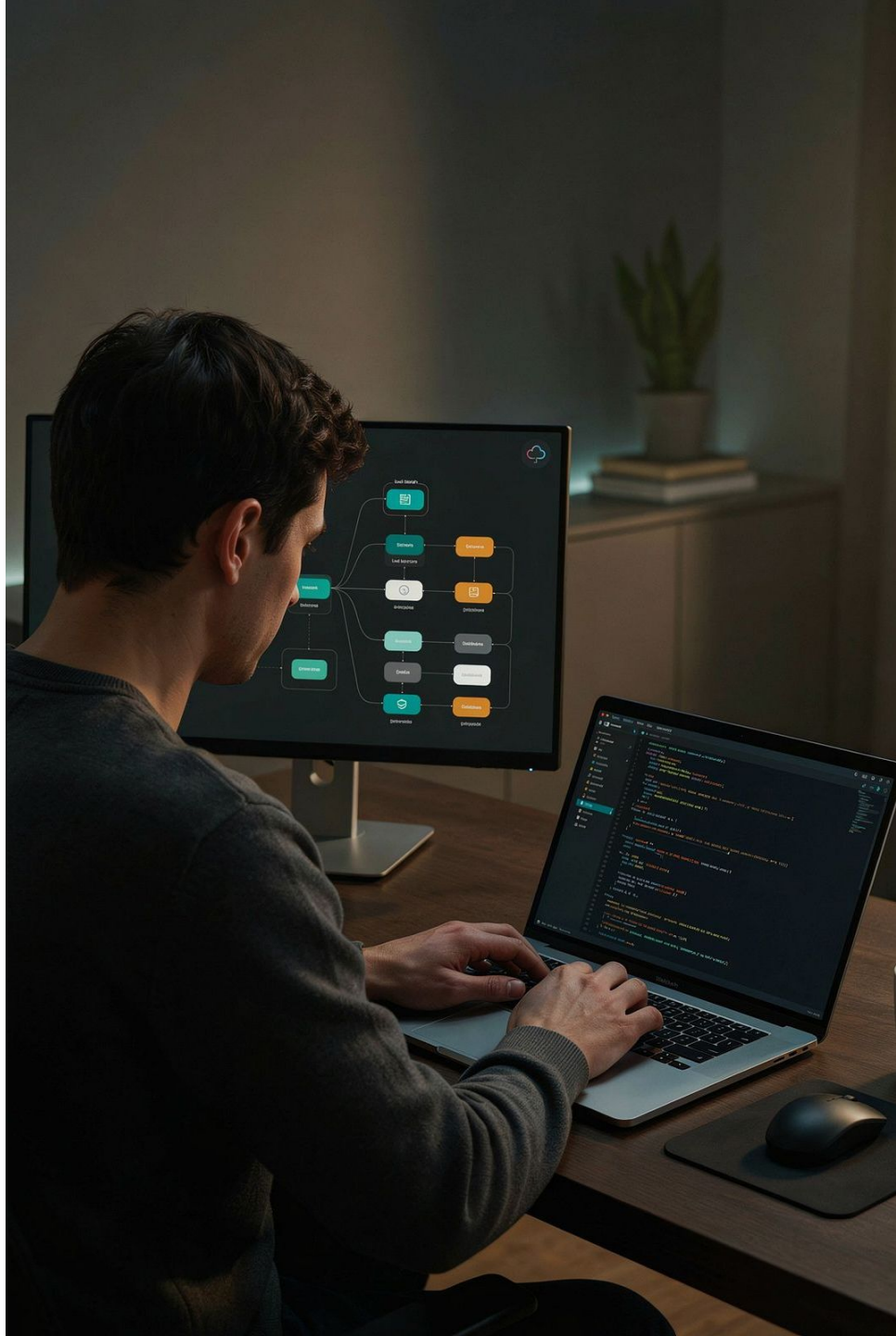
AWS Resource Explorer

Search and discover resources across all regions instantly

TOOL 1

Amazon Q Developer for IaC

Infrastructure as Code (IaC) is powerful — but writing CloudFormation YAML or Terraform HCL from scratch is time-consuming and error-prone. **Amazon Q Developer** acts like a coding co-pilot that generates, explains, and refactors your IaC templates in plain English.



AMAZON Q FOR IAC

Real-World Example: Launch a VPC in Minutes

What You Say

"Create a CloudFormation template for a VPC with two public subnets, two private subnets, an internet gateway, and a NAT gateway."

That's it. No manual YAML. No hunting through documentation for property names.

What Q Delivers

- A complete, deployable CloudFormation or Terraform template
- Inline comments explaining each resource block
- Security best-practice defaults baked in
- Suggestions for parameterizing environment-specific values

What used to take an experienced engineer 45 minutes now takes under 2 minutes.

TOOL 2

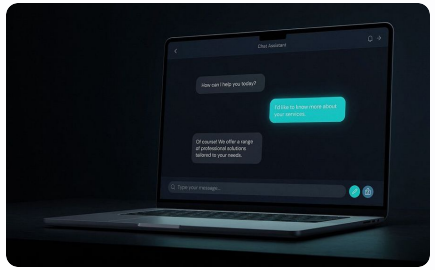
AWS Console Q Assistant

The AWS Console has hundreds of services and thousands of settings. Even experienced engineers forget where things live. The **Console Q Assistant** is a chat panel built directly into the AWS Management Console — ask it anything about AWS and get instant, contextual answers without leaving the page.



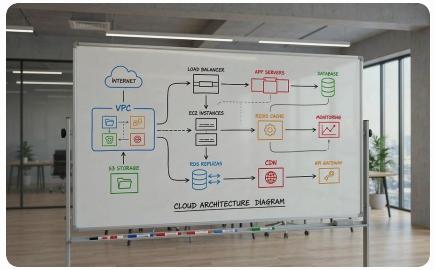
Ask in Plain English

"How do I enable versioning on my S3 bucket?" — Q explains the steps and can navigate you directly there.



Context-Aware Guidance

Q knows which service page you're on and tailors answers accordingly — no need to re-explain your context.



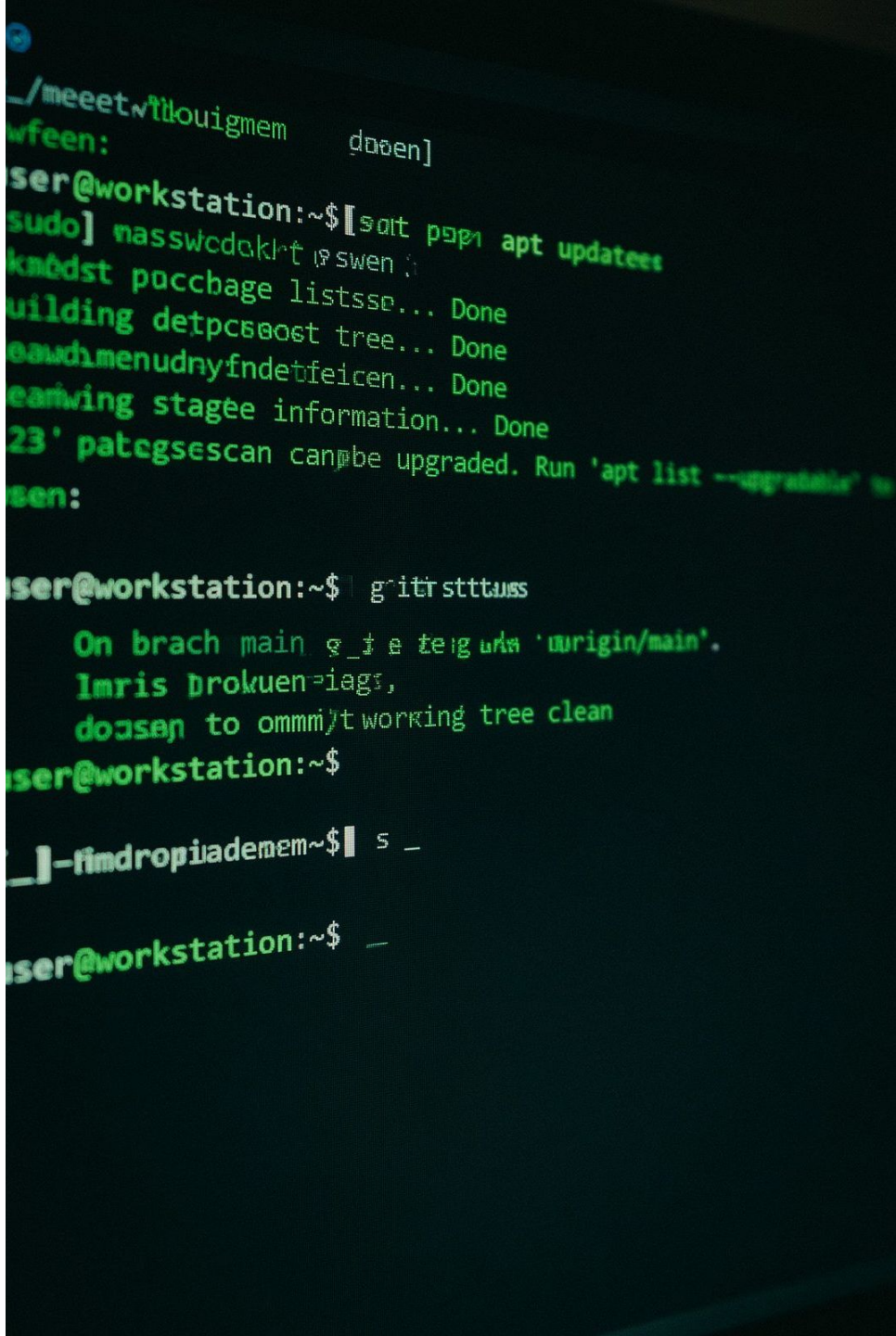
Service Recommendations

"What's the best way to cache my RDS queries?" — Q suggests ElastiCache and explains when to use Redis vs. Memcached.

TOOL 3

Amazon Q in the CLI

The command line is where engineers live — but memorizing every AWS CLI flag and syntax is unrealistic. **Amazon Q integrates directly into your terminal**, offering intelligent auto-complete, command translation from natural language, and inline explanations so you stay in flow without reaching for a browser tab.



CLI WITH Q

Stop Googling CLI Commands

Before Q in the CLI

- Open a browser, search "aws s3 sync command example"
- Copy a Stack Overflow answer and hope it works
- Spend 10 minutes tweaking flags and debugging errors
- Repeat for every unfamiliar command

After Q in the CLI

- Type: **"sync my local folder to S3 and delete files not in source"**
- Q auto-generates the exact command with the right flags
- Explains what --delete does before you run it
- Suggests safer alternatives if the command is destructive

Real example: Ask Q "list all EC2 instances that are running in us-east-1 and show their instance type" — it writes the full CLI command instantly.

TOOL 4

CloudWatch Logs Insights

When something breaks in production at 2 AM, the last thing you want to do is wrestle with complex query syntax. **CloudWatch Logs Insights** lets you run powerful, structured queries across your application logs — and with Amazon Q assistance, you can describe what you're looking for in plain English and get query suggestions instantly.

🔍 Find Errors Fast

Filter logs by error code, status, or keyword across millions of log lines in seconds — no grep required.

📊 Aggregate & Visualize

Count error occurrences by minute, calculate p99 latency, or group by service — all with a single query.

🤖 AI-Assisted Queries

Describe your goal in plain language and Q helps translate it into the correct Logs Insights syntax.

TOOL 5

AWS Resource Explorer

AWS Resource Explorer is your unified search engine for everything you've ever deployed on AWS. As teams grow and environments multiply — dev, staging, production, multiple regions — it becomes genuinely difficult to know what's running where. Resource Explorer solves this with a single, cross-region, cross-service search interface.



AWS RESOURCE EXPLORER

Find Any Resource, Anywhere, Instantly

The Problem It Solves

Imagine your team has 8 AWS accounts, 4 active regions, and 200+ resources deployed over 3 years. Someone asks: "Do we have an RDS instance in eu-west-1?" Without Resource Explorer, you're clicking through the console region by region.

With Resource Explorer

- Search by resource type, tag, region, or name in one place
- Works across EC2, S3, Lambda, RDS, IAM, and more
- Filter by tag to find all "production" or "team-A" resources
- Identify orphaned or forgotten resources to reduce costs

